

Zehui Yin

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🆔 0000-0001-6954-7918 🎓 [IC7pmh0AAAAJ&hl](#) 🌐 [zehuiyin.github.io](#) | Updated: 2026-Aug

Education

McMaster University, PhD in Geography 9/2024 – Present
School of Earth, Environment & Society Hamilton, Ontario, Canada
Supervisor: [Dr. Darren Scott](#)

University of Toronto Scarborough, HBA 9/2020 – 6/2024
Major in Economics for Management Studies Toronto, Ontario, Canada
Minor in Geographic Information Science & Applied Statistics
Certificate in Computational Social Science
Graduated with High Distinction
Cumulative GPA: 3.98 out of 4

Honours & Awards

Kanaroglou Memorial Graduate Scholarship 2025, 2026
C\$1,500 scholarship awarded to a graduate student specializing in transportation research within the School of Earth, Environment & Society

Esri Canada Center of Excellence Student of the Year 2025
C\$750 scholarship awarded to students who excel in GIS and demonstrate the mission and vision of the center

Ontario Graduate Scholarship 2025
C\$15,000 total merit-based scholarship to a graduate student for three consecutive terms jointly funded by the Government of Ontario and McMaster University

Graduation Prize in Management and Economics 2024
C\$250 award for the top student graduating from the Department of Management and Economics

Governor General's Silver Medal 2024
Highest academic standing upon graduation from a bachelor's degree program
There are three silver medals awarded at the University of Toronto

University of Toronto Excellence Awards 2023, 2024
C\$7,500 scholarship for undergraduate students to conduct summer research projects

Department of Human Geography Outstanding Academic Performance Award 2024

Highest graduating cumulative GPA in the Minor in Geographic Information Systems

Vincent Bladen In-course Scholarships 2023

C\$300 scholarship for exceptional academic achievement

Sotherton Wadhams In-course Scholar 2023

C\$1,500 scholarship for high academic standing

University of Toronto Scarborough Dean's List 2021, 2022, 2023

Annual GPA equal to or greater than 3.5 in the previous academic year

University of Toronto Scholar 2021

C\$1,500 scholarship for top 20 students in the previous academic year

Professional Experience

Research Assistant, *TransLAB, McMaster University* 9/2024 – Present

Project 1: Hamilton Shared Micromobility (Bike Share and Shared E-scooter) Data

Project 2: "Future Mobility in Canada Survey"

Project 3: "Transportation Tomorrow Survey"

Supervised by: [Dr. Darren Scott](#)

Research Assistant, *Suburban Mobilities Cluster, University of Toronto Scarborough* 11/2022 – 8/2024

Project 1: "Newcomers' Accessibility to Agencies in Scarborough"

Project 2: "Metrolinx Service for OSAP Students at UTSC"

Project 3: "Towards sustainable neighborhoods? Tensions and heterogeneous transport priorities among suburban residents"

Project 4: "Evaluating the transit accessibility and equity of the Bus Rapid Transit (BRT) system: The case of Dar-es-Salaam, Tanzania"

Project 5: "Toronto Cycling Survey"

Project 6: "Neighbourhood Satisfaction in Scarborough: A Discrete Choice Analysis with Spatial Effects"

Project 7: "Cycling evolution in Toronto (1999–2023)"

Supervised by: [Dr. Steven Farber](#), [Dr. Andre Cire](#), and [Dr. Ignacio Tiznado-Aitken](#)

Teaching Experience

Instructor

ENVSOCY 3GV3 Advanced Vector GIS, *McMaster University* Fall 2026

Sessional Faculty for an upper-year undergraduate GIS course

Course covers vector data modelling, geodatabase design, network analysis, location analytics, geocoding, and automated workflows with ModelBuilder

Design and deliver weekly lectures; develop and coordinate hands-on GIS assignments using ArcGIS Pro and ArcGIS Business Analyst; develop and administer written term tests; supervise teaching assistants

Guest Lecturer

ENVSOCY 3MA3 Research Methods, *McMaster University*

March 14, 2025

Presentation title: Introduction to Quantitative Analysis in Research

Teaching Assistant

ENVSOCY 4GA3 Applied Spatial Statistics,
McMaster University

Winter 2025, Winter & Fall 2026

Teaching Assistant for [Dr. Antonio Paez](#) (Winter 2025, Fall 2026) and Bruno Dos Santos (Winter 2026)

Conducted weekly lab sessions for two groups of approximately 24 students each

Held office hours and provided additional support via email

Graded assignments, midterm and final reflections

ENVSOCY 4GS3 GIS Programming, *McMaster University*

Fall 2025

Teaching Assistant for [Dr. Darren Scott](#)

Graded assignments and responded to student inquiries via email

ENVSOCY 4GT3 Web GIS, *McMaster University*

Fall 2025

Teaching Assistant for [Patrick DeLuca](#)

Graded assignments

ENVSOCY 2GI3 Geographic Information Systems, *McMaster University*

Fall 2024

Teaching Assistant for [Dr. Darren Scott](#)

Conducted weekly lab sessions for two groups of approximately 25 students each

Invigilated examinations and graded assignments and tests

GGR30H3 Geographic Information Systems (GIS) and Empirical Reasoning, *University of Toronto Scarborough*

Summer 2023

Teaching Assistant for [Dr. Shaila Jamal](#)

Conducted weekly lab sessions for around 30 students

Held weekly office hours to provide additional support

Invigilated examinations and graded assignments and tests

Publications

Tiznado-Aitken, I., **Yin, Z.**, & Farber, S. (2026). Tracking cycling behaviour and comfort perceptions in Toronto: Trends from 1999 to 2023. *Transportation Research Part A: Policy and Practice*, 210, 105031. <https://doi.org/10.1016/j.tra.2026.105031>

- Yin, Z., & Scott, D. M.** (2026). Wheels within reach: Evaluating bike share equity impacts of a pseudo-free student pass using network-distance-based kernel density. *Journal of Transport Geography*, 134, 104666. <https://doi.org/10.1016/j.jtrangeo.2026.104666>
- Yin, Z., & Scott, D. M.** (2026). Pedal preferences: GPS-based panel data insights into bike share traffic flow across membership groups. *Cities*, 170, 106703. <https://doi.org/10.1016/j.cities.2025.106703>
- Yin, Z., & Scott, D. M.** (2026). Decoding the first-and-last mile: Analyzing the transit connecting bike share GPS routes in Hamilton, Ontario with spatiotemporal distance decay. *Travel Behaviour and Society*, 42, 101122. <https://doi.org/10.1016/j.tbs.2025.101122>
- Yin, Z., Jamal, S., Tiznado-Aitken, I., & Farber, S.** (2025). What makes people satisfied with their neighbourhoods? Exploring individual characteristics beyond sociodemographics in Scarborough, Ontario. *Applied Geography*, 183, 103738. <https://doi.org/10.1016/j.apgeog.2025.103738>
- Tiznado-Aitken, I., **Yin, Z.**, & Farber, S. (2025). Towards sustainable neighbourhoods? Tensions and heterogeneous transport priorities among suburban residents. *Transportation Research Part D: Transport and Environment*, 138, 104514. <https://doi.org/10.1016/j.trd.2024.104514>
- Mwesigwa, L., **Yin, Z.**, & Farber, S. (2024). Evaluating the level of access and equity of the bus rapid transit (BRT) system: The case of Dar-Es-Salaam, Tanzania. *Journal of Transport Geography*, 119, 103953. <https://doi.org/10.1016/j.jtrangeo.2024.103953>
- Yin, Z.**, Rybarczyk, G., Zheng, A., Su, L., Sun, B., & Yan, X. (2024). Shared micromobility as a first- and last-mile transit solution? Spatiotemporal insights from a novel dataset. *Journal of Transport Geography*, 114, 103778. <https://doi.org/10.1016/j.jtrangeo.2023.103778>
- Huang, E., **Yin, Z.**, Broaddus, A., & Yan, X. (2024). Shared e-scooters as a last-mile transit solution? Travel behavior insights from Los Angeles and Washington D.C. *Travel Behaviour and Society*, 34, 100663. <https://doi.org/10.1016/j.tbs.2023.100663>

Reports and Other Publications

- Forrister, A., Yan, X., **Yin, Z.**, Zhao, X., Cova, T., Lovreglio, R., Nilsson, D., & Kuligowski, E. (2022). Survey of Evacuation Behavior in the 2021 Marshall Fire, Colorado. *Natural Hazards Center Quick Response Grant Report Series*, 349. Natural Hazards Center, University of Colorado Boulder. <https://hazards.colorado.edu/quick-response-report/survey-of-evacuation-behavior-in-the-2021-marshall-fire-colorado>

Work in Progress

- E-bikes in Bikeshare: Longitudinal Evidence from San Francisco and Washington, D.C., with Duanya Lyu, Darren Scott, Liang Zhai, and Xiang Yan
- Modeling E-Bike Route Choice with Street-Level Imagery and Infrastructure Attributes: A Path Size Logit Analysis in Washington, DC, with Yiheng Qian, Darren Scott, and Xiang Yan

Conference Presentations

(Note: Presenters are underlined.)

Qian, Y., Yin, Z., Scott, D. M., Yan, X. (2026). Modeling e-bike route choice with street-level imagery and infrastructure attributes: A path size logit analysis in Washington, DC. *Transportation Research Board 2026 Annual Meeting*, Washington D.C.

Lyu, D., Yin, Z., Scott, D. M., Zhai, L., & Yan, X. (2026). What e-bikes bring to bikeshare: Evidence from a longitudinal analysis of bikeshare systems in San Francisco and Washington, DC. *Transportation Research Board 2026 Annual Meeting*, Washington D.C.

Yin, Z., & Scott, D. M. (2025). Decoding the first-and-last mile: Analyzing the transit connecting bike share GPS routes in Hamilton, Ontario with spatiotemporal distance decay. *ITE Toronto-Hamilton-Southwest Ontario Sections – Mini Transportation Conference*, Oakville, Ontario.

Yin, Z., Rybarczyk, G., Zheng, A., Su, L., Sun, B., & Yan, X. (2024). Shared micromobility as a last-mile transit solution? Spatiotemporal insights from a novel dataset. *Inaugural U.S. Department of Transportation Future of Transportation Summit*, Washington D.C.

Yin, Z., Rybarczyk, G., Zheng, A., Su, L., Sun, B., & Yan, X. (2024). Shared micromobility as a first- and last-mile transit solution? Insights from a novel dataset. *Transportation Research Board 2024 Annual Meeting*, Washington D.C.

Mwesigwa, L., Yin, Z., & Farber, S. (2024). Evaluating the Transit Accessibility and Equity of the Bus Rapid Transit (BRT) System: The Case of Dar-Es-Salaam, Tanzania. *Transportation Research Board 2024 Annual Meeting*, Washington D.C.

Tiznado-Aitken, I., Yin, Z., & Farber, S. (2023). Towards sustainable neighborhoods? Heterogeneous transport priorities among suburban Scarborough residents. *NECTAR Cluster 6-7 Workshop – Sustainable neighborhoods: Urban and transport planning for sustainable urban living*, Coimbra, Portugal.

Huang, E., Yin, Z., Broaddus, A., & Yan, X. (2023). Transit and shared e-scooter integration: Travel behavior insights from Los Angeles and Washington D.C. *Transportation Research Board 2023 Annual Meeting*, Washington D.C.

Invited Talks

“Pedal Preferences: GPS-Based Panel Data Insights into Bike Share Traffic Flow Across Membership Groups.” *Geographic Information Science (GIS) Day, McMaster University*, Hamilton, Ontario. November 19, 2025.

“Decoding the First-and-Last Mile: Analyzing the Transit Connecting Bike Share GPS Routes in Hamilton, Ontario with Spatiotemporal Distance Decay.” *School of Earth, Environment & Society Graduate Research Symposium, McMaster University*, Hamilton, Ontario. April 11, 2025.

“What Makes People Happy with Their Neighbourhoods? Exploring Individual Characteristics Beyond Sociodemographics in Scarborough, Ontario.” *Geographic Information Science (GIS) Day, McMaster University*, Hamilton, Ontario. November 21, 2024.

“What Makes People Happy with Their Neighbourhoods? Exploring Individual Characteristics Beyond Sociodemographics in Scarborough, Ontario.” *Suburban Mobilities Research Day, University of Toronto*. Toronto, Ontario. November 14, 2024. <https://www.mobilitynetwork.utoronto.ca/suburban-mobilities-research-day>

“Toward Equitable Service Provision: A Catchment Area Analysis of the Bike Share Toronto system.” *Civic Hacknight #441, Civic Tech Toronto*. Toronto, Ontario. April 30, 2024. <https://civictech.ca/hacknights/441>

“Investment Priority on Transportation Infrastructures.” *Suburban Mobilities Cluster Day, University of Toronto*. Toronto, Ontario. June 9, 2023. <https://www.mobilitynetwork.utoronto.ca/suburban-mobilities-cluster-at-utsc-celebrates-its-first-research-day-with-a-focus-on-improving-transportation-at-scarborough>

Departmental Service

Student Associate, <i>Esri Canada GIS Centers of Excellence at McMaster</i>	2024 – Present
Member, <i>School of Earth, Environment & Society School Council</i>	2024 – 2025
Geography Representative, <i>School of Earth, Environment & Society Graduate Student Collective</i>	2024 – 2025

University Service

Member, <i>Social Science Student Research Ethics Committee (SocSciSREC), McMaster University</i>	2024 – Present
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Professional Service

Manuscript Reviewer (Note: Number of completed reviews indicated in parentheses.)

Transportation Research Part A (1), *Transportation Research Part D* (3), *Cities* (1), *Journal of Transport Geography* (3), *Travel Behaviour and Society* (3), *Transport Policy* (1), *Transportation* (2), *Case Studies on Transport Policy* (2), *Transportation Research Record* (1), *Transportation Research Board Annual Meeting* (4), *Journal of Cycling and Micromobility Research* (1), *Scientific Reports* (1), *Humanities and Social Sciences Communications* (1), *EPJ Data Science* (1), *Journal of Urban Technology* (1)

Languages

English: Full professional proficiency

Mandarin Chinese: Native proficiency

Research Interests

Spatial Analysis, Econometrics, Machine Learning, Transportation Planning, Public Transit, and Micromobility

Programming Languages & Software

Proficient: R (Spatial/GIS: *sf*, *terra*; Modeling: *apollo*, *tidymodels*; Visualization: *ggplot2*, *plotly*, *shiny*), Python (Spatial/GIS: *GeoPandas*, *Rasterio*, *Xarray*, *rioxarray*, *ArcPy*; Modeling: *statsmodels*, *scikit-learn*, *PyTorch*; Visualization: *Matplotlib*, *seaborn*), ArcGIS Pro, ArcGIS Online, Markdown (R Markdown, Quarto), Microsoft Office Suite

Working Experience: QGIS, L^AT_EX, Overleaf, HTML, CSS, JavaScript, Leaflet, Turf.js, MapLibre GL JS, Mapbox GL JS, SQL, Git, GitHub

References

Available upon request